USER & OPERATING MANUAL: DRIER SIZING TENSION CONTROLLER

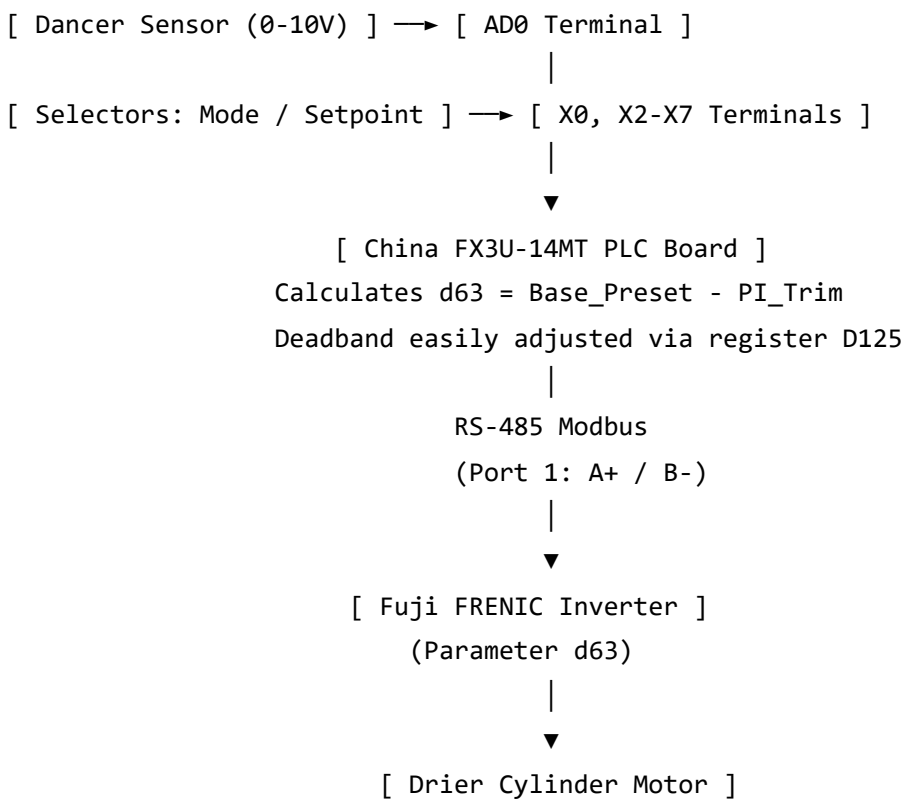
China FX3U-14MT-6AD-2DA PLC & Fuji Electric FRENIC Inverter Integration

1. System Overview

This control system synchronizes and regulates the drying cylinder (*drier*) speed on a **Textile Warp Sizing Machine**.

The system operates **100% automatically** using a closed-loop **Proportional-Integral (PI)** algorithm:

1. It reads continuous yarn tension from an analog dancer roll (**AD0** : 0 – 10V DC → 0 – 4000 kg).
2. It compares the tension against the active tension setpoint (X0 =1000kg, X5 =1500kg, X6 =2000kg, X7 =2500kg).
3. It filters the tension through an **easily adjustable deadband register (D125)** to prevent mechanical hunting.
4. It dynamically calculates the required speed ratio correction (**Parameter d63**).
5. It transmits the updated d63 value directly to a **Fuji Electric Inverter** via **RS-485 Modbus RTU**.



2. Hardware Specifications

Item	Specification
PLC Model	China FX3U-14MT-6AD-2DA (STM32F103 ARM Cortex-M3 @ 72 MHz)
Power Supply	**DC 24V** ($\pm 10\%$), 1 Amp regulated power supply
Digital Inputs	8 Channels (`X0 – X7`), NPN Low-Level (Active LOW / Sinking)
Digital Outputs	6 Channels (`Y0 – Y5`), Transistor NPN Open Collector (DC 24V, 500mA max)
Analog Inputs	6 Channels (`AD0 – AD5`), 12-bit resolution (`0 to 4095` counts)
Analog Used	**`AD0`** : Voltage `0 – 10V DC` connected to Dancer (scaled to `0 – 4000 kg`)
Communication	**COM1:** RS232 Programming/HMI **COM2:** RS-485 Terminals A/B (Modbus RTU Master)
Target Drive	Fuji Electric FRENIC Series (MEGA / Ace / Multi / Mini)

3. Terminal Wiring Diagram

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================================================================================
                              TOP TERMINAL STRIP
================================================================================
[ 24V ] ──── Connect to External +24V DC Power Supply (+)
[  0V ] ──── Connect to External  0V DC Power Supply (-)

--- DIGITAL INPUTS (NPN: Connect terminal to 0V / GND to activate) ---
[ X0  ] ──── Setpoint Switch: SETPOINT 1000 kg (Dedicated Target 1000 kg)
[ X1  ] ──── Spare (Unused)
[ X2  ] ──── Machine Mode:    SIPUT FLAG       (Selects Base d63 = 2375 via D205)
[ X3  ] ──── Machine Mode:    SPEED 1 FLAG     (Selects Base d63 = 2300 via D210)
[ X4  ] ──── Machine Mode:    SPEED 2 FLAG     (Selects Base d63 = 2225 via D215)
[ X5  ] ──── Setpoint Switch: SETPOINT 1500 kg (Dedicated Target 1500 kg)
[ X6  ] ──── Setpoint Switch: SETPOINT 2000 kg (Dedicated Target 2000 kg)
[ X7  ] ──── Setpoint Switch: SETPOINT 2500 kg (Dedicated Target 2500 kg)

--- ANALOG INPUTS ---
[ AD0 ] ──── Dancer Analog Signal (+) (0 to 10V DC)
[ AD1 ] ──── Spare (0-10V)
[ AD2 ] ──── Spare (0-10V)
[ AD3 ] ──── Spare (0-20mA)
[ AD4 ] ──── Spare (0-20mA)
[ AD5 ] ──── Spare (0-20mA)
[AGND ] ──── Dancer Analog Signal Ground Reference (-)

================================================================================
                             BOTTOM TERMINAL STRIP
================================================================================
--- TRANSISTOR OUTPUTS (DC Loads Only) ---
[ Y0  ] ──── Spare
[ Y1  ] ──── Spare
[ Y2  ] ──── Spare
[ Y3  ] ──── Spare
[ Y4  ] ──── Spare
[ Y5  ] ──── Spare
[COM0 ] ──── Connect to DC 0V (GND)

--- ANALOG OUTPUTS ---
[ DA0 ] ──── Spare Analog Output 1 (0-10V)
[ DA1 ] ──── Spare Analog Output 2 (0-10V)
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[AGND ] ——— Analog Ground

--- RS-485 SERIAL PORT ---
[ A+  ] (or D+) ——— Connect to Fuji Inverter RS-485 Terminal (+) / Pin 4 (RJ45)
[ B-  ] (or D-) ——— Connect to Fuji Inverter RS-485 Terminal (-) / Pin 5 (RJ45)
================================================================================
```

> ⚠ **Caution**
>
> **Analog Ground vs. Power Ground:** Always connect the dancer sensor's ground wire to `AGND`, never to the digital `0V` terminal. This prevents electrical switching noise from corrupting the tension signal.

4. Fuji Inverter Communication Configuration

Ensure the following parameters are configured on the Fuji Inverter keypad:

Parameter	Setting Name	Value	Description
`y01 / y11`	Station Address	`1`	Modbus Slave Station ID = 1
`y02 / y12`	Comms Error Action	`0`	Trip or coast to stop on communication loss
`y03 / y13`	Comms Error Timer	`1.0`	1.0 second communication error timeout
`y04 / y14`	Baud Rate	`2`	**9600 bps** (matches PLC register `D8120`)
`y05 / y15`	Data Length	`0`	**8 Data bits**
`y06 / y16`	Parity	`1`	**Even Parity**
`y07 / y17`	Stop Bits	`0`	**1 Stop bit**
`y10 / y20`	Protocol Selection	`0`	**Modbus RTU Protocol**
`d62`	Pulse Scaling 1	*(Fixed)*	Pulse numerator (set per machine gearing)
`d63`	Pulse Scaling 2	*(Auto)*	**Controlled dynamically by PLC via Modbus**

5. Control Philosophy & Operating Principles

A. The Sizing Machine Mechanical Dynamics

$$\text{Inverter Frequency } (f^*) = \text{Master Pulses } (Np) \times \left(\frac{\mathbf{d62}}{\mathbf{d63}} \right)$$

Because the dancer roll is positioned **AFTER the drier cylinders** (between Drier and Beamer):

- The **Drier FEEDS yarn** into the dancer.
- The **Beamer PULLS yarn** out of the dancer.

$$\text{Drier Speed } \propto \frac{1}{\mathbf{d63}}$$

- **When Tension is TOO HIGH ($> \text{Setpoint} + \mathbf{D125}$):**
  - The beamer is pulling faster than the drier feeds. Yarn is being stretched.
  - The drier must **FEED MORE YARN (SPEED UP)**.
  - To speed up the drier, the PLC **DECREASES** `d63` via positive PI Trim.
- **When Tension is TOO LOW ($< \text{Setpoint} - \mathbf{D125}$):**
  - The drier is feeding faster than the beamer pulls. Yarn is sagging.
  - The drier must **FEED LESS YARN (SLOW DOWN)**.
  - To slow down the drier, the PLC **INCREASES** `d63` via negative PI Trim.
- **Within Deadband ($\text{Setpoint} \pm \mathbf{D125}$):**
  - Tension is inside the target zone. The PI controller holds steady. **No adjustments are made.**

B. Easy Deadband Adjustment (`D125`)

The deadband is stored in **Register** `D125` (Default = `100 kg`):

- The PLC automatically calculates the negative limit in `D126` (`0 - D125 = -100`).
- **How to change on-site:** Anyone with GX Works2, an HMI, or handheld programmer can simply open register monitor and write a new value into `D125` (e.g. `50` , `80` , `120`) **in real time without editing or compiling ladder code!**

C. Dedicated Speed Modes and Base Presets

$$\mathbf{d63} = \text{Active_Base_Preset} - \mathbf{PI_Trim}$$

The 3 machine speed modes are dedicated to `X002`, `X003`, and `X004`:

1. `SIPUT` **Mode (** `X002` **):** Base `d63 = 2375` (Saved in `D205`). Lowest speed for crawl/threading.
2. `SPEED 1` **Mode (** `X003` **):** Base `d63 = 2300` (Saved in `D210`). Normal production speed.
3. `SPEED 2` **Mode (** `X004` **):** Base `d63 = 2225` (Saved in `D215`). High-speed production.

The **PI Trim (** `D150` **)** continuously floats on top of whichever mode is selected:

$$\text{PI_Trim} = \text{P_Term} + \text{I_Accumulator}$$

- **P-Term:** Instantly reacts to tension displacement.
- **I-Accumulator:** Gently accumulates every 200 ms to eliminate long-term tension offset.

D. Dedicated Tension Setpoint Selection (`X0`, `X5`, `X6`, `X7`)

Terminal	Target Tension Setpoint	Register `D120` Value
`X000`	**1000 kg**	`D120 = 1000` (Also power-up default)
`X005`	**1500 kg**	`D120 = 1500`
`X006`	**2000 kg**	`D120 = 2000`
`X007`	**2500 kg**	`D120 = 2500`

E. Built-In Safeguards

1. **VFD EEPROM Memory Protection:**
   The PLC compares current `d63` with the last sent value (`LD<> D230 D231`). Modbus transmission is triggered **only when the value changes**, preventing the inverter's memory chip from wearing out.
2. **Web-Break / Line-Stop Lockout:**
   If the yarn snaps or the line stops, the dancer drops to 0 kg.
   The PLC checks if `Tension > 150 kg`. If tension is below 150 kg, the PI controller freezes and resets `Trim = 0` so the drier will **never run away to high speed**.

3. **Integral Anti-Windup Clamping:**
   The integral accumulator is hard-clamped between `-80` **and** `+80` counts, preventing massive overshoots when accelerating or starting the machine.

6. Complete Memory Register Reference

Device	Name / Function	Range / Default	Description
X000	`SETPOINT 1000KG`	Digital Input	Dedicated 1000 kg tension selector
X001	Spare	—	Unused
X002	`SIPUT FLAG`	Digital Input	Dedicated Siput mode selector (Base `d63 = 2375`)
X003	`SPEED 1 FLAG`	Digital Input	Dedicated Speed 1 mode selector (Base `d63 = 2300`)
X004	`SPEED 2 FLAG`	Digital Input	Dedicated Speed 2 mode selector (Base `d63 = 2225`)
X005	`SETPOINT 1500KG`	Digital Input	Dedicated 1500 kg tension selector
X006	`SETPOINT 2000KG`	Digital Input	Dedicated 2000 kg tension selector
X007	`SETPOINT 2500KG`	Digital Input	Dedicated 2500 kg tension selector
M2	Siput Flag	Internal Relay	Active when in Siput mode
M3	Speed 1 Flag	Internal Relay	Active when in Speed 1 mode
M4	Speed 2 Flag	Internal Relay	Active when in Speed 2 mode
M50	Modbus Trigger	Internal Relay	Pulses ON when new `d63` needs transmission
T20	PI Sampling Timer	200 ms	Controls the integration clock rate
D100	Raw ADC Count	0 to 4095	Raw 12-bit analog input from terminal `AD0`
D104	Unfiltered Tension	0 to 4000 kg	Raw tension calculated from `AD0`

Device	Name / Function	Range / Default	Description
`D110`	Filtered Tension (PV)	0 to 4000 kg	Clean 4-sample running average of dancer tension
`D120`	Tension Setpoint (SV)	1000 – 2500 kg	Selected tension target
`D122`	Raw Error	$\mathrm{PV} - \mathrm{SV}$	Deviation from setpoint
`D124`	Effective Error	Error after Deadband	Error fed into PI algorithm (0 inside deadband)
`D125`	**Deadband Setting**	**100 kg**	**Tuning register: easily adjust deadband live!**
`D126`	Negative Deadband	Calculated (`-D125`)	Automatic negative threshold (`0 - D125`)
`D128`	P Product 32-bit	—	Intermediate multiplication register
`D130`	Proportional Term	-80 to $+80$	$(\mathrm{Error} \times K_p)/100$
`D132`	I Product 32-bit	—	Intermediate multiplication register
`D134`	Integral Increment	—	Delta step per 200ms
`D140`	Integral Sum	-80 to $+80$	Accumulated steady-state correction
`D150`	Total Dynamic Trim	-100 to $+100$	Sum of $P + I$
`D205`	Base Siput `d63`	2375 (`H947`)	Base ratio for Siput mode
`D210`	Base Speed 1 `d63`	2300 (`H8FC`)	Base ratio for Speed 1 mode
`D215`	Base Speed 2 `d63`	2225 (`H8B1`)	Base ratio for Speed 2 mode
`D220`	Active Base `d63`	2000 – 3000	The base ratio for currently active mode
`D230`	Final `d63` Output	$\mathrm{Base} - \mathrm{Trim}$	**Value written to Fuji Inverter Parameter `d63`**
`D231`	Last Sent `d63`	2000 – 3000	History register to detect value changes
`D320`	Proportional Gain K_p	**5**	Tuning register for immediate response

Device	Name / Function	Range / Default	Description
`D322`	Integral Gain K_i	**2**	Tuning register for steady-state offset removal
`D8120`	RS-485 Config	`H8087`	9600 bps, 8 Data, Even Parity, 1 Stop bit
`D8129`	Modbus Timeout	500 ms	Serial communication timeout

7. How to Load the Program into the PLC

Using Mitsubishi GX Works2:

1. Open **GX Works2**.
2. Go to **Project** → **New...**
   - **Series:** `FXCPU`
   - **Type:** `FX3U / FX3UC`
   - **Project Type:** `Simple Project` (Ladder)
3. Connect your PC to the PLC's programming port using a USB-to-RS232 programming cable (COM port typically set to **38.4 kbps**, 7-E-1).
4. Go to **Project** → **Open Other Format Data** → **Import from Text File...**
   - Select: `export/MAIN export.txt`
5. Go to **Edit** → **Documentation** → **Import from Text File...**
   - Select: `export/COMMENT export.txt`
6. Press `F4` (Compile / Rebuild All).
7. Go to **Online** → **Write to PLC...**, check **Program** and **Comment**, and click **Execute**.
8. Turn the PLC RUN/STOP switch to **RUN**. The `RUN` LED should glow solid green.

8. On-Site Tuning Guide

You can monitor and adjust the controller directly from GX Works2 Online Monitor (`F3`):

1. Adjusting the Deadband (`D125`):

- **Default:** `D125 = 100` (± 100 kg).

- If you want tighter tension control with narrower tolerance → Change **`D125`** to **`50`** (± 50 kg) or **`70`** (± 70 kg).
- If the yarn has natural slight tension wobble and you want less control activity → Change **`D125`** to **`120`** or **`150`**.

2. Adjusting Gains (`D320` and `D322`):

DANCER RESPONSE BEHAVIOR	WHAT TO ADJUST
The dancer reacts too slowly to line speed changes, and tension wanders outside the deadband for too long:	⟶ Increase Kp (D320) from 5 to 7 or 8.
The dancer is too aggressive, starts bouncing or vibrating, and causes the drier speed to oscillate:	⟶ Decrease Kp (D320) from 5 to 3 or 4.
The dancer settles smoothly, but stays offset just outside the deadband without returning to center:	⟶ Increase Ki (D322) from 2 to 3 or 4.
Tension overshoots the setpoint when returning to center:	⟶ Decrease Ki (D322) from 2 to 1.

9. Troubleshooting & Diagnostics

Symptom	Probable Cause	Action
`ERR` LED is flashing on PLC	Hardware watchdog or syntax fault	Check DC 24V power supply voltage. Ensure memory range settings in GX Works2 match FX3U.
Dancer reads `0 kg` (`D100 = 0`)	Wiring broken or 0-10V signal missing	Measure DC voltage between terminal `AD0` and `AGND` with a multimeter. Ensure sensor power supply is ON.

Symptom	Probable Cause	Action
Dancer reads `4000 kg` constantly	Open ground wire or input saturation	Check `AGND` terminal connection. Ensure input signal does not exceed 10.0V DC.
Inverter does not change speed	RS-485 wiring reversed or baud mismatch	1. Swap terminals `A+` and `B-` on RS-485 port. 2. Verify Inverter parameters `y01 = 1`, `y04 = 2` (9600 bps), `y06 = 1` (Even), `y10 = 0` (Modbus).
Inverter trips on Comms Fault (`ErP`)	Timeout threshold too strict	Check inverter parameter `y03` (set to `1.0` second or higher). Ensure cable shielding is grounded at one end.
Tension increases when drier accelerates	Inverted dancer physical placement	If dancer was physically moved *before* the drier instead of after, change `SUB D220 D150 D230` to `ADD D220 D150 D230` in Section 7.